

Geometric Programs for Solenoidal Forcing

OpenAI

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Abstract

We construct smooth solenoidal mean-zero forces, periodic from time zero, for which a fixed particle on a flat three-torus reaches a fixed open strip exactly when a given machine halts. The initial fluid velocity and the pressure are zero. We also realize positive diagonal maps on coding sheets by incompressible shears, with explicit normal compensation for changes of planar area, and reciprocal maps on whole boxes of positive thickness. Complete local inverses, initialization rules, and intermediate trajectory bounds connect these geometric constructions to finite computations.

1 The geometry of a finite fluid instruction

A finite list of symbolic instructions becomes a fluid construction only after several geometric questions have been answered. Does an instruction act on every point near a code? Can its target overlap another source? How is the input supplied without changing the requested time symmetry? Does a particle avoid the detector during the motion between two successive codes? These questions distinguish constructions that have the same integer-time symbolic action.

We study finite lists of affine maps between closed rectangles, together with the local tape rules that produce those maps. Throughout, a rectangle is an axis-parallel Cartesian product of intervals in the stated coordinates. A *geometric instruction* specifies a source rectangle, a target rectangle, and an affine bijection between them. Sources in a list must be mutually separated, and so must targets; intersections between the two lists are allowed. A reciprocal diagonal instruction expands one coordinate by the factor by which it contracts the other. A general positive diagonal instruction may change the rectangle's area. We realize these operations on sheets or on boxes of positive thickness using smooth transverse velocity pulses, with explicit control of their entire trajectories.

1.1 Main results

Write $\mathbb{T}_L^3 = (\mathbb{R}/L\mathbb{Z})^3$ for the flat torus. At a fixed viscosity $\nu > 0$ we use

$$u_t + (u \cdot \nabla)u = -\nabla p + \nu \Delta u + f, \quad \operatorname{div} u = 0, \quad u(0, \cdot) = 0. \quad (1)$$

Pressure is spatially periodic and normalized to have mean zero. A classical comparison solution has continuous u, u_t , spatial derivatives of u through order two, and $p, \nabla p$ on every finite closed time cylinder. The material flow is defined by $\partial_t X(t, a) = u(t, X(t, a))$, $X(0, a) = a$. An effective force prescription evaluates every requested mixed derivative at computably supplied space and time arguments to any positive rational error. The construction also supplies effective global bounds when these are asserted. Its finite formula describes all instruction branches; it is not a recording of an executed computation.

Theorem 1.1 (Periodic initialization). *Fix a computable viscosity $\nu > 0$. From a deterministic one-tape machine, with head moves in $\{-1, 0, 1\}$, and a finite input, one can effectively construct a smooth force on $\mathbb{T}_1^3$ such that:*

- (i) *The force is divergence free, has zero spatial mean, and is one-periodic from time zero. Equation (1) has a global smooth solution with pressure zero, unique in the stated classical class.*
- (ii) *The velocity is one-periodic, vanishes near every integer time, has zero instantaneous self-advection, and has uniformly bounded kinetic energy. Every mixed derivative of force and velocity is globally bounded.*
- (iii) *For the fixed label $a_* = (1/4, 1/2, 1/4)$ and fixed open strip $O = \{1/2 < x_1 < 7/8\}$, the material trajectory enters O at a finite real time exactly when the machine halts, including an initially halted machine.*

The construction may use one Cartesian velocity component at a time. For arbitrary fixed real $\nu > 0$ the same formulas hold and are effective relative to that viscosity.

Section 4 proves this theorem after the common toolkit in Section 3. It gives two initialization mechanisms: an input-dependent chart and a reserved loading branch in the repeated table. The latter gives the one-component option. Both incorporate initialization into a schedule that repeats from time zero. Alternative recorder guards and observers are developed separately in Section 5.

The second result allows an instruction to change planar area. Its normal action then changes as well, as volume preservation requires.

Theorem 1.2 (General diagonal maps on a sheet). *Let $P_i, Q_i \subset [2, 3]^2$, $1 \leq i \leq N$, be closed rectangles with rational vertices and positive side lengths. Suppose the sources are pairwise positively separated and the targets are pairwise positively separated. Let $F_i : P_i \rightarrow Q_i$ be the center-to-center affine map with any positive rational diagonal linear part, with $F_i(P_i) = Q_i$. For each computable $\nu > 0$ there is an effective smooth mean-zero solenoidal force on $\mathbb{T}_{10}^3$, one-periodic from time zero, whose unique solution in the classical class has pressure zero and period map*

$$(y, 2) \mapsto (F_i(y), 2) \quad (y \in P_i).$$

Its velocity is zero near integer times and has zero self-advection; all mixed derivatives of force and velocity are bounded. No condition is imposed on source–target intersections. For $N = 0$ use zero.

The sequential storage construction in Section 8.1 proves this theorem for arbitrary such rectangle data; Section 8.2 applies it to a complete tape computation. Section 8.4 gives a simultaneous construction for the narrow-strip rectangles of a different recorder, using separate heights: contraction, translation, and expansion take fifteen phases. Both proofs exhibit the compensating normal action instead of treating an area-changing planar map as area preserving. If the required action is on a whole box of positive thickness, Section 7.1 provides a separate reciprocal box-transfer theorem with an all-time sign guard.

1.2 How the constructions fit together

A finite local recorder stores enough information to reverse each rule on its entire partial domain. Two positional coordinates encode the tape halves. Fixing the finitely many letters read by a rule gives a closed rectangle, and the inverse ensures that the image rectangles are separated. The toolkit in Section 3 makes this passage explicit.

Each reciprocal rectangle map is then performed at a private height. A vertical pulse lifts the source rectangle; four centered planar shears produce the diagonal change of shape; a translation moves it to its target center; a final vertical pulse returns it to the coding height. Separating these stages in time makes each active field transverse, so convection vanishes. Spatial selectors have zero mean and are constant on the tracked paths. Thus the direct force formula gives the fluid equation, while the partial shear paths give the observation bounds. Sections 6 and 7 adapt this mechanism to a one-time loader, other clocks and charts, and whole boxes.

For arbitrary planar area change, the normal direction must also move. A vertical shear records a horizontal offset in height, a horizontal shear changes that offset, and a second vertical shear restores the sheet while rescaling nearby normal offsets. To allow a target to intersect any source, the general construction first evacuates all source sheets into a separate storage region. It then delivers them one at a time. This geometry proves the second theorem before any symbolic application is chosen; its low-height bound supplies the application’s fixed detector.

1.3 Ideas and relation to earlier work

An inverse cannot discard the information that distinguishes predecessors. Landauer discusses this retention principle and its storage implications [5, Section 3]. Reversible Turing machines already appear in Lecerf’s work on undecidability for code isomorphisms [6]. Bennett’s reversible simulation records each instruction and verifies that the local rules have disjoint domains and disjoint ranges [1, Eqs. (5)–(11) and Table 1]. Our finite local recorders use these principles; their particular guards are checked on the full partial domains, including tapes outside the initialized run.

Moore’s generalized shifts connect positional tape coding to dynamical systems [8]. They admit finite affine Cantor-block representations, and invertible generalized shifts have smooth realizations [9, Lemma 0 and Section 6]. Our geometric constructions implement the resulting closed-rectangle maps with explicit transverse pulses, mean-zero forcing, and observation bounds throughout each operation. Separation of sources and of images is the interface between the symbolic inverse and these pulse formulas.

The fluid setting also has substantial predecessors. Cardona, Miranda, Peralta-Salas and Presas realize computation by stationary Euler flows on an adapted Riemannian three-sphere [3]. Their Proposition 5.1 constructs an area-preserving realization by contracting, transporting, and expanding neighborhoods with a relative area correction. That geometric organization is related to the sheet construction below. For a fixed Euclidean metric, Cardona, Miranda and Peralta-Salas construct Turing-complete stationary Euler fields on $\mathbb{R}^3$; their simulation uses a noncompact set and the fields have infinite energy [2]. On the standard flat three-torus they robustly simulate tape-bounded machines, with reachability tested before a trajectory leaves a prescribed compact region. Their construction also gives time-dependent Navier–Stokes realizations of these bounded computations [2].

Dyhr, González-Prieto, Miranda and Peralta-Salas obtain stationary unforced viscous computation after a metric deformation, using the Hodge Laplacian on one-forms as the viscous operator [4, Theorem A and Proposition 3.1]. Here the flat metric, the usual flat Laplacian, positive viscosity, and zero initial velocity are fixed; the finite machine and input specify the external force. The local shear formulas supply the required geometry directly.

The companion *Finite Instructions and Solenoidal Shear Flows* [10] gives one complete initialized construction and the common reciprocal-rectangle theorem. We restate its geometric realization (Theorem 3.2) in Section 2, and its recorder and extensions (Lemma 2.1 and Section 5.1) in Section 6.3. The present proofs address changed requirements: periodicity from zero, alternative full-domain guards, routing through a common center, positive thickness, and planar area change. A

symbolic inverse is used to prove separation; it does not replace the geometric or intermediate-time argument.

At each instant our active velocity moves only in directions on which its coefficients do not depend. Consequently convection vanishes and $f = U_t - \nu \Delta U$ gives solenoidal forcing with pressure zero. Section 2.1 proves the relevant uniqueness statement by the classical difference-energy method associated with Leray [7, Section 18]. Slower onto clocks preserve the complete trajectory, including initialization; their exact derivative estimates are proved in Section 6.1. A clock with finite accumulated time would not suffice.

Finally Section 9 writes the same equation in material labels, and Appendix A constructs a fixed universal table in the stated tape model. Thus the force compiler can either take an arbitrary finite machine table or keep one interpreter fixed and vary only its input. Composing Theorem 1.1 with any decision procedure for its fixed particle event would decide halting. This includes Turing’s symbol-printing question: stop when the designated symbol is printed, and otherwise continue forever, including when the original machine stops without printing it [11, Section 8]. The constructions use exact real codes and assert no uniform finite-precision tolerance.

2 From transverse pulses to fluid motion

2.1 Transverse pulses and classical uniqueness

A transverse field has either one nonzero component independent of its direction of motion, or the form $(a(z), b(z), 0)$. Both its divergence and its self-advection vanish. A finite sum of such fields need not have zero self-advection if several act simultaneously; we instead give them disjoint time supports. This scheduling restriction is part of every construction below.

Lemma 2.1 (Direct solenoidal forcing). *Let $U(t, x)$ on a flat torus be a smooth sequence of transverse pulses with disjoint temporal supports, each of zero spatial mean, and let $U(0) = 0$. Assume U and its derivatives are bounded on each finite time interval. Then $f = U_t - \nu \Delta U$ is smooth, solenoidal and mean zero, and $(U, 0)$ solves (1). Its velocity and normalized pressure are unique in the classical comparison class defined above. Its material flow exists for every finite forward time and is a volume-preserving smooth diffeomorphism. If a finite pulse sequence is repeated, with zero collars at its joins, all mixed derivatives are globally bounded and the energy is uniformly bounded. These conclusions also hold when one separate finite loading sequence is prepended.*

Proof. At each time at most one transverse field is active, so $\operatorname{div} U = 0$ and $(U \cdot \nabla)U = 0$. Divergence commutes with ∂_t and Δ , and a periodic Laplacian has zero integral. These facts prove the force assertions and the equation by substitution. For another classical solution v , put $w = v - U$. Subtract the equations, multiply by w , and integrate on the torus. Incompressibility and periodic integration by parts cancel the pressure and transport by v , giving

$$\frac{1}{2} \frac{d}{dt} \|w\|_2^2 + \nu \|\nabla w\|_2^2 \leq \|\nabla U\|_{\infty, \text{op}} \|w\|_2^2.$$

The stated classical regularity justifies each operation. The coefficient is bounded on every finite interval and $w(0) = 0$, so an integrating factor gives $w = 0$. The pressure gradient then vanishes, and its normalization makes the pressure zero. This is a comparison for the displayed global solution, not an existence assertion for general data.

The spatial periodic lift of U is bounded and Lipschitz on every finite interval. Local ODE existence, uniqueness and continuation give its trajectories, and backward integration supplies their inverses. Smooth dependence gives smooth diffeomorphisms. The Jacobian obeys $\partial_t \det D_a X =$

$(\operatorname{div} U) \circ X \det D_a X$ with initial value one, proving volume preservation. A finite loading interval followed by a fixed smooth repeated pattern has a compact collection of phase templates; their derivative bounds imply all the last assertions. $\square$

2.2 The reciprocal geometric input

Theorem 3.2 of [10] has the following precise interface. Fix rational $L, h > 0$ and rational coordinate charts on $\mathbb{T}_L^3$. Take finite families of closed axis-parallel rational rectangles P_i, Q_i with positive side lengths and centers p_i, q_i , each family pairwise positively separated, and maps $F_i(y) = q_i + \operatorname{diag}(r_i, r_i^{-1})(y - p_i)$ with $F_i(P_i) = Q_i$ and $r_i > 0$ rational. Suppose every source and target half-width is at most h , their centers lie in a rational rectangle K , and $K + [-2h, 2h]^2$ is compactly contained in the planar chart. For any rational coding height z_0 inside its chart, that theorem supplies seven effective mean-zero transverse pulses, periodic with zero collars, whose period map is $(y, z) \mapsto (F_i(y), z)$ on a positive effectively specified neighborhood of each $P_i \times \{z_0\}$. For a point starting on P_i at height z_0 , the four deformation stages remain within sup distance $2h$ of the source center; the following translation interpolates between centers with a fixed offset of sup norm at most h . Lifting and lowering preserve the planar coordinates. Source–target intersections are allowed. The theorem also proves the empty-list case and the force conclusions of Lemma 2.1.

This seven-stage construction supplies the reciprocal template used below. Alternative shear lists and translation routes are checked locally, including their cutoff plateaus and intermediate paths. The constructions on boxes of prescribed thickness likewise prove their own bounds on every point of those boxes.

3 A common geometric and symbolic toolkit

The constructions share two interfaces: transverse pulses realize finite rectangle maps, and local inverse rules make the rectangle families separated. We first specify the profiles and paths used by the pulses, then the positional coding that converts a finite rule table into their input. The recorder itself is introduced in Section 4.

3.1 Profiles that preserve the exact shear paths

Write $X(a)(x, y) = (x + ay, y)$ and $Y(a)(x, y) = (x, y + ax)$. Besides the four shears in Theorem 3.2 of [10], we will use these chronological lists about a branch center:

$$Y(-r), X(r^{-1} - 1), Y(1), X(r - 1), \quad (2)$$

$$Y(-r(r - 1)), X(-r^{-1}), Y(r - 1), X(1), \quad (3)$$

$$X(r^{-1} - 1), Y(1), X(r - 1), Y(-r^{-1}). \quad (4)$$

All have product $\operatorname{diag}(r, r^{-1})$, with products taken in reverse chronological order. For example, the leftmost three factors of the full product in (3) multiply to $\begin{pmatrix} r & 0 \\ r^{-1} & r^{-1} \end{pmatrix}$; right multiplication by $Y(-r(r - 1))$ finishes the identity. For (4) the rightmost three factors are $\begin{pmatrix} r & 0 \\ 1 & r^{-1} \end{pmatrix}$, which the final shear makes diagonal.

If $r, r^{-1} \leq B$ and initial half-sides are at most $\ell/2$, the crude bounds for every partial shear are $(1 + B)^4 \ell/2$ for (2),(4), and $(1 + B^2)^4 \ell/2$ for (3). Each follows by four applications of $\|X(a)\|_\infty, \|Y(a)\|_\infty \leq 1 + |a|$; partial pulse coefficients have smaller modulus. For (2), the stronger endpoint-size estimate of Theorem 3.2 of [10] bounds offsets by $3\ell/2$ whenever both the source and target sides are at most ℓ , independently of r . These estimates are useful for different chart choices.

Here are three interchangeable ways to impose zero mean while retaining the specified paths. A box cutoff always equals one on a neighborhood of the indicated closed box, and is supported in a rational collar. For an explicit convention set $E(s) = e^{-1/s}$ for $s > 0$ and zero otherwise, and $H(s) = E(s)/(E(s) + E(1 - s))$. For $I = [a, b]$ use

$$\chi_{I,\delta}(s) = H(2(s - a + \delta)/\delta)H(2(b + \delta - s)/\delta).$$

It is one on $[a - \delta/2, b + \delta/2]$, supported in $[a - \delta, b + \delta]$, and symmetric about the interval center. Products give box cutoffs. Positive-side derivatives of E are polynomials in $1/s$ times $e^{-1/s}$, and the bound $e^{-1/s} \leq n!s^n$ gives effective flatness at zero to every order. The denominator of H is at least e^{-2} . Thus these profiles and all their derivatives can be evaluated without deciding equality with a cutoff endpoint.

1. A rectangle selector may be $\partial_x((x - c_x)\chi_P(x, y))$, and a height selector may be $\partial_z((z - z_i)\zeta_i(z))$. They equal one on their plateaus and have integral zero by periodic differentiation.
2. A selector may be $\chi_P(x, y) - \chi_P(x, y - \delta)$, or $\zeta_i(z) - \zeta_i(z - \delta)$, when the translated support misses all tracked rectangles or heights. Equal translated integrals cancel. A mean-zero height selector makes every supported horizontal linear profile have zero spatial mean, even if that linear profile does not itself have zero mean.
3. With an even local cutoff, the periodic local function $\Xi_c(s) = (s - c)H(2 - 2(s - c)^2/D^2)$ is odd about c , hence has zero mean. Its derivative also has zero mean. On $|s - c| \leq D/2$ they equal $s - c$ and 1, respectively. Alternatively, $\partial_s(\frac{1}{2}(s - c)^2\chi(s))$ and $\partial_s(s\chi(s))$ give the same linear and constant plateaus.

All profiles are extended periodically only across a region where they vanish. The first recipe requires a neighborhood plateau because a derivative is taken; the second only requires the explicit separation from translated supports.

3.2 The seven-stage construction and its pulse convention

The reciprocal theorem in Section 2 uses a source selector to assign each rectangle its own height, performs its planar map there, and uses a target selector to return it to the coding height. We write the fields from its proof [10, Theorem 3.2] in a form allowing separate shear and translation selectors, since later constructions change individual profiles or stages.

Write the centers of P_i, Q_i as c_i^-, c_i^+ and the prescribed linear part as $\text{diag}(r_i, r_i^{-1})$. Let z_0 be the coding height. Choose smooth periodic selectors A_i^-, A_i^+ with zero planar mean, equal to one near their own source or target rectangle and zero near all other rectangles of the same family. Choose distinct private heights z_i . Two height selectors σ_i, τ_i equal one near z_i and zero near every other private height. Require τ_i to have zero mean. For $j = 1, 2$, choose a periodic profile g_{ij} equal to $s - c_{ij}^-$ on a neighborhood of the coordinate range of the four centered shear paths, and require

$$\left(\int \sigma_i\right)\left(\int g_{ij}\right) = 0.$$

Thus the shear selector σ_i may have nonzero mean when both linear profiles have zero mean. Usually one can take $\sigma_i = \tau_i$. The derivative, odd-profile, and translated-copy recipes above provide these choices; their compensating parts must miss the other tracked rectangles or private-height plateaus as specified there.

For the chronological shear list (2), the seven spatial fields are

$$\begin{aligned}
W_1 &= e_z \sum_i (z_i - z_0) A_i^-(x, y), \\
W_2 &= e_y \sum_i (-r_i) \sigma_i(z) g_{i1}(x), & W_3 &= e_x \sum_i (r_i^{-1} - 1) \sigma_i(z) g_{i2}(y), \\
W_4 &= e_y \sum_i \sigma_i(z) g_{i1}(x), & W_5 &= e_x \sum_i (r_i - 1) \sigma_i(z) g_{i2}(y), \\
W_6 &= \sum_i \tau_i(z) (c_i^+ - c_i^-, 0), \\
W_7 &= e_z \sum_i (z_0 - z_i) A_i^+(x, y).
\end{aligned} \tag{5}$$

At each stage all branch contributions have the same direction of motion, except in W_6 , whose two horizontal components depend only on height. Consequently each entire stage is transverse, even where profile supports overlap. The stated integral conditions give zero spatial mean to every summand and hence to each W_j .

To turn the stages into smooth motion, choose successive closed rational intervals $[a_j, b_j]$ with $0 < a_j < b_j < 1$ and positive gaps, and set

$$\beta_j(t) = \frac{1}{b_j - a_j} H' \left(\frac{t - a_j}{b_j - a_j} \right), \quad U(t, x) = \sum_j \beta_j(t) W_j(x) \quad (0 \leq t \leq 1). \tag{6}$$

The step function H defined above is nondecreasing, so each pulse is nonnegative, has integral one, and vanishes to every order at its endpoints. The finite list repeats smoothly with period one and zero collars at integer times. This convention applies to any finite number of repeated stages below. The same pulse formula is used on a separately specified finite loading interval. Its derivatives and their bounds are effective by the same flat-profile estimates. Distinct stages have disjoint temporal supports; branch contributions within one stage act simultaneously. Thus the summed velocity still has zero self-advection, and Lemma 2.1 applies.

Here is the exact path represented by these fields. Starting from (y, z_0) with $y \in P_i$, W_1 lifts the point to (y, z_i) . Only the i th height selectors act during the next five stages. The four shears send the offset $y - c_i^-$ to $\text{diag}(r_i, r_i^{-1})(y - c_i^-)$, still based at c_i^- . The sixth stage translates that center to c_i^+ , and W_7 lowers the resulting point of Q_i back to z_0 . Nonnegative pulse masses parameterize each partial shear or translation by a number in $[0, 1]$. The excursion estimates above therefore verify the linear plateaus on the full closed source rectangle before ODE uniqueness identifies these formulas with the actual trajectories. Positive plateau margins also permit a sufficiently small neighborhood of that rectangle and coding height; its size must be chosen from the margins and the affine partial maps. The bounds for points on the original rectangle are not automatically bounds for that neighborhood.

The other four-shear lists replace $W_2, \dots, W_5$ by their listed coefficients and directions, using their own excursion estimates. Splitting W_6 into an x -translation and a y -translation gives the eight-stage version. If extra transverse controls are used, the first is one on the current y -interval and the second on the target x -interval already reached; both retain the private-height selector. Their scalar coefficients must have zero integral, supplied by either factor or by a correction supported off every tracked path. Remote centers and routes with an extra translation are specified separately below: their endpoint maps use the same principles, while their intermediate paths require the bounds proved for those routes.

3.3 A digit interface for full rule domains

A relative tape is a bi-infinite sequence $T = (T_j)_{j \in \mathbb{Z}}$ whose index zero is the current head position. If an edit produces $\tilde{T}$, displacement d gives $T'_j = \tilde{T}_{j+d}$. A full cylinder fixes a mode and finitely many tape letters, with every other letter arbitrary. This convention specifies the domains on which we will prove local inverses.

Let an alphabet of size m have distinct digits separated by at least two in base B , with all digits in $\{0, \dots, B-2\}$. Write

$$C(a_1 a_2 \dots) = \sum_{j \geq 1} d(a_j) B^{-j}, \quad I(v) = \left[\sum_{j=1}^{|v|} d(v_j) B^{-j}, \sum_{j=1}^{|v|} d(v_j) B^{-j} + B^{-|v|} \right].$$

The two coordinates of a relative tape are $C(T_{-1} T_{-2} \dots)$ and $C(T_0 T_1 \dots)$. Place the modes in pairwise positively separated translated squares of common side length ℓ , and use these two codes as the coordinates within each square. This is the positional-coding principle of Moore [9]; the following lemma records the full-rectangle and separation properties used here.

Lemma 3.1 (Full cylinders give full rectangles). *Suppose a finite injective partial rule map is partitioned into full cylinders specified by a mode and the tape window $[-a, b-1]$. A branch edits only that window and shifts by d , with $a+d, b-d \geq 0$. Its image cylinder has the rewritten prefixes of lengths $a+d, b-d$. The associated closed source and target rectangles are separately pairwise separated. The exact branch map on the whole rectangle has linear part $\text{diag}(B^{-d}, B^d)$.*

Proof. A prefix v followed by an arbitrary tail has code $C(v) + B^{-|v|} C(\text{tail})$. The two free tails survive the edit and shift, so changing prefix lengths gives the stated affine factors on all tail parameters in $[0, 1]$. Thus each image really is a full cylinder. If two cylinders in one mode are disjoint, their left prefixes or their right prefixes must be incompatible; otherwise arbitrary compatible extensions would give a common tape. At the first differing digit of incompatible words, their intervals have a positive gap, at least the corresponding digit scale. This proves source separation and, by injectivity, target separation. Distinct mode squares are separated by their placement. A constant tail has rational code $B^{-k} d(a)/(B-1)$ after a prefix of length k ; therefore all finite input codes are rational and effectively known. Positive gaps allow successive finite-prefix decoding from promised codes, including codes on rectangle boundaries. $\square$

For a single-cell table there is a shorter sufficient inverse test: the target mode determines the incoming displacement, and the target mode together with the written letter determines the old mode and old letter. One first reverses the displacement, then the unique write. Let $T_a(v) = (d(a) + v)/B$ and $I_a = T_a([0, 1])$. Reading a , writing b , and moving by d gives the following complete rectangle formulas:

d	source	(L', R')	target
1	$[0, 1] \times I_a$	$(T_b(L), T_a^{-1}(R))$	$I_b \times [0, 1]$
0	$[0, 1] \times I_a$	$(L, T_b T_a^{-1}(R))$	$[0, 1] \times I_b$
-1	$I_c \times I_a$	$(T_c^{-1}(L), T_c T_b T_a^{-1}(R))$	$[0, 1] \times T_c(I_b)$

(7)

Include the last row for every c . Determinism separates sources. The inverse test separates target written letters; in the last row the pair (c, b) separates the two-digit target intervals. If mode-square gaps are at least ℓ , all within-family gaps are at least ℓ/B^2 in one coordinate.

Common force conclusions. In each construction below choose nonnegative smooth unit-integral pulses strictly inside the listed phases and repeat their finite list with period one. The velocity is zero near integer times, so the initial velocity is zero. Every fixed mixed derivative of the force and velocity is bounded, and the kinetic energy is uniformly bounded. The force is effective for computable $\nu > 0$, and the same formula is effective relative to an arbitrary supplied positive viscosity. Unless stated otherwise, m is the augmented tape alphabet size, N the recorder mode count, J the number of rectangle branches, and (L_*, R_*) the rational initialized left and right codes.

4 A compiler periodic from time zero

We now prove Theorem 1.1. The recorder below retains each instruction before moving the simulated head. Its full-domain inverse provides separated rectangle maps; two ways of incorporating the input then make the entire pulse schedule periodic from time zero. A separate one-time loader would give only eventual periodicity.

Throughout the recorder constructions a machine instruction indexed by $i = (q, a)$ has data (q_i, b_i, d_i) , where $d_i \in \{-1, 0, 1\}$. Missing instructions may be completed by unchanged-symbol, zero-displacement moves to a halt state. Every table below is partial: an unlisted case has no rule. On initialized tapes, all auxiliary tracks are blank away from the specified finite initialization, and all unspecified mark bits are zero. The full partial domains are restricted only by the displayed local guards.

When an idle nonhalting start is requested, add a fresh state with one unchanged-symbol, zero-displacement instruction for each tape letter, leading to the original start state. No instruction targets the fresh state. This adds one simulated step, preserves halting even when the original start is already halted, and leaves the initial tape unchanged.

4.1 Recording before the simulated move

This recorder marks the old head, logs the instruction, and only then makes the simulated move. Let $e \in \{-1, 0, 1\}$ remember the preceding move, let $r = (q, e, a)$, and give its instruction the data (q_r, b_r, d_r) . Letters are (a, h, m) , with $h \in \{E, F\} \sqcup \{r\}$ and $m \in \{0, 1\}$. The controls are $M_{q,e}, R_r, P_{q,d}, L_{q,d}$. The following table uses $h \neq F$:

source	read	written	move	target
$M_{q,e}$	$(a, h, 0)$	$(b_r, h, 1)$	1	R_r
R_r	$(c, h, 0)$	$(c, h, 0)$	1	R_r
R_r	$(c, F, 0)$	$(c, r, 0)$	1	P_{q_r, d_r}
$P_{q,d}$	$(c, E, 0)$	$(c, F, 0)$	-1	$L_{q,d}$
$L_{q,d}$	$(c, h, 0)$	$(c, h, 0)$	-1	$L_{q,d}$
$L_{q,d}$	$(c, h, 1)$	$(c, h, 0)$	d	$M_{q,d}$

(8)

Only nonhalting q have a first row. Initialize the frontier at absolute cell c_0 , where either $c_0 = 1$ or $c_0 = 2$, and start in $M_{q_0,0}$. At the n th main visit the head is the machine head h_n , the records occupy $c_0, \dots, c_0 + n - 1$, and the frontier is $J_n = c_0 + n$. Since $h_n \leq n < J_n$, the first row is defined. It marks h_n , the right scan logs at J_n , the placement row moves the frontier to $J_n + 1$, and the left scan returns to the unique mark. The last row makes the original move. This is exactly $2(J_n - h_n) + 3$ local steps, a finite positive number; for $c_0 = 2$ it lies between 7 and $4n + 7$. Thus main halts and original halts agree, with no intervening main visit.

This table also passes the inverse test on arbitrary tapes. Into R_r , the written mark distinguishes entry from its loop and r recovers the old work symbol and control. Into P , the written record

identifies the predecessor. Into L , a written frontier distinguishes placement from scanning. Into $M_{q,d}$, the sole predecessor restores a mark and has known displacement d . All other tracks are retained. Incoming displacements are respectively $1, 1, -1, d$, proving the test.

The choices $c_0 = 1$ and $c_0 = 2$ have the same entire rule table but different initialized histories. A permanent origin bit may be added and copied by every rule. When $c_0 = 1$ without such a bit, the leftmost nonempty history cell remains absolute cell one and also determines absolute indexing. These are precise ways of recovering the head position, rather than additional assumptions on arbitrary tapes.

An old-head recorder with an input-shifted chart. Use (8) with frontier at two and an idle nonhalt start. Put $\ell = 1/(64N(1+B)^4)$, with odd digits in base $B = 2m + 1$. The initial origin is $(1/4 - \ell L_*, 1/2 - \ell R_*)$; other nonhalt origins are $(1/4 + 2j\ell, 1/2)$ for $j \geq 1$, and halt origins are $(5/8 + 2j\ell, 1/2)$ with a fresh enumeration. All squares remain separated, and nonhalt centers have first coordinate at most $9/32$. Use collars $\ell/(10B^2)$, selectors corrected by a $1/4$ shift in y , and heights $1/2 + i/(8(J+1))$ with collar $1/(32(J+1))$, corrected by a $1/4$ shift in z . The seven phases with (2) have excursion $1/(128N)$ inside $[1/8, 7/8]$. Nonhalt paths have $x < 1/2$; halt endpoints lie in $[5/8, 21/32]$. The label is $(1/4, 1/2, 1/4)$ and the fixed event is $1/2 < x < 7/8$.

There is a second exact specialization of the same rule table: put the frontier at one, use odd digits in base $B = 2m + 2$, set $\ell = 1/(64(N+1))$, and use the same initial-chart shift and other origins. Use derivative rectangle and height masks and the endpoint-size bound $3\ell/2$, which does not grow with B . The label remains $(1/4, 1/2, 1/4)$ and the event can be narrowed to $1/2 < x < 3/4$. Indeed nonhalt paths satisfy $x \leq 1/4 + 2N\ell + 3\ell/2 < 1/2$, while halt endpoints lie in $[5/8, 5/8 + 2N\ell] \subset (1/2, 3/4)$. This preserves the different initialized history and observer without repeating the six-rule inverse proof.

A reserved loading target. Instead make a fresh copy q_0 of the original start state, preserving its halting status, and arrange that no instruction targets this copy. In (8), retain placement and return controls only for state-displacement pairs that occur as instruction targets. No rule then enters $M_{q_0,0}$. For N remaining modes set $\ell = 1/(32(N+1)(1+B)^4)$ and put their origins at $(1/4 + 2j\ell, 1/4)$ or $(5/8 + 2j\ell, 1/4)$ according to halting status. The initial code q^{in} has coordinate margins at least $\ell/(B-1)$ inside its mode square. Add a translation from the square centered at $(1/4, 1/2)$ with half-width $\ell/(4(B-1))$ to the congruent square at q^{in} . Its source is separated in y ; its target is inside a square having no incoming rule. Thus both augmented families are separated. Use eight phases with (2), derivative rectangle masks, and local profiles Ξ_c with $D = 1/16$. The excursion $(1+B)^4\ell/2 < 1/32$ fits every plateau; use a collar at most one quarter of $\min(1/32, \ell/B^2, 1/(4(J+1)))$. For nonhalting loading and later transitions, $x \leq 5/16 + (1+B)^4\ell/2 < 1/2$; halt targets lie in $[5/8, 11/16]$. The label and event are again $(1/4, 1/2, 1/4)$ and $1/2 < x < 7/8$. Initially halted inputs hit during the first loading period. The loader remains in the repeated table, so periodicity begins at zero.

Completion of Theorem 1.1. The input-shifted chart gives all the stated force and event properties. The reserved loading branch gives the same fixed label and observer, with eight axial phases and hence one Cartesian component at a time. Both satisfy Lemma 2.1; their rational parameters and effective flat profiles give the stated derivative evaluation. $\square$

5 Alternative recorder contracts

The headline compiler is complete. The alternatives in this section change what information is present at initialization or at a return to an ordinary machine state. Their local guards matter: each inverse must hold on every allowed tape, including tapes outside the initialized run. Instruction histories retain the information needed for reversibility [1]; the rules below specify how that information is recovered in each case.

Section 5.1 writes a finite input from an entirely blank tape inside the repeated table. Section 5.2 trades a neighbor guard for shorter history updates and then keeps a persistent head mark. Section 5.3 uses that mark for a reserved loader and an exact map on a box of positive thickness. Section 5.4 combines the work and departure steps while retaining an origin flag. Each rule table is followed by its inverse, its initialized simulation, and a geometric realization with a fixed observer; these are different guarantees rather than extra steps in the proof of Theorem 1.1.

5.1 Writing the input inside the repeated rule table

A one-time loader gives eventual periodicity. To keep periodicity from time zero, the loading operation can instead be a distinguished local branch. Here is a version starting from an entirely blank tape. Let $r = (q, e)$, with $e \in \{-1, 0, 1, \star\}$; only initialization uses $\star$. For a nonhalt instruction put $k = (r, a)$, $s(k) = (q', d)$. Letters are (a, m, g) with $g \in \{E, P\} \sqcup \{k\}$; modes are I, C_r, S_k, B_s .

1. In I require cells $0, \dots, R_0 - 1$ fully blank, where $R_0 = \max(2, |w|)$. Write the input and a history pointer at cell one, and enter $C_{(q_0, \star)}$ without moving.
2. In nonhalt C_r require $m_0 = 0, g_0 \neq P$; write the instruction symbol, mark cell zero, move right into S_k .
3. In S_k require $m_0 = 0$. Scan right if $g_0 \neq P$; if $g_0 = P, g_1 = E$, write k, P , move left into $B_{s(k)}$.
4. In B_s require $g_1 \neq P$. At $m_0 = 0$ scan left; at $m_0 = 1$ erase the mark, move by the displacement in s , and enter C_s .

The star tag distinguishes initialization from every other arrival in a primary mode; it restores exactly the required blank block. At an S_k output, the mark at -1 distinguishes entry and scanning. At a B_s output, the pointer at two distinguishes turning and scanning; the record at one identifies k . An ordinary primary output has a specified return state. These are full-domain inverse tests. At the main visit after n steps, the frontier is $n + 1 > h_n$ and the records occupy $1, \dots, n$. The old-head shuttle therefore takes $2(n - h_n) + 3$ rules, including the final erase-and-move rule. An initially halting machine is reached immediately after the initializer. The ordinary rules are resolved on $[-1, 1]$, the initializer on $[0, R_0 - 1]$; Lemma 3.1 applies to both.

A blank initial tape and an initialization branch. Use Section 5.1. Give the fully blank symbol digit zero, the other symbols the remaining even digits, and use $B = 2m + 1$. Set $\ell = 1/(16N(1+B^2)^4)$ and enumerate modes with I first. Their origins are $(1/4, 1/4 + 2j\ell)$, except that primary halt modes have first coordinate $5/8$. The initial blank code is $(1/4, 1/4, 1/4)$. Derivative rectangle masks and the primitive-derived linear and constant profiles above implement eight axial phases with (3). All centered excursions are at most $1/(32N)$ and lie within $[1/8, 7/8]^2$. If the target is nonterminal, $x \leq 1/4 + \ell + 1/(32N) < 1/2$ at every phase; halt targets have $x \in [5/8, 5/8 + \ell] \subset (1/2, 3/4)$. The initializer's star tag proves that its target family is disjoint from every ordinary arrival. Thus the fixed event is $1/2 < x < 3/4$, including immediate halting after the initialization rule.

5.2 Neighbor guards and a persistent head

Here the history append writes two adjacent cells at once. This saves a local step, but the inverse now needs a neighbor guard.

An erased return mark. Letters are (a, m, g) , with $g \in \{E, P\} \sqcup \{i\}$; controls are A_q, B_i, C_q . Use the following complete rules:

1. At nonhalt A_q , require $m_{d_i} = 0$, write $a_0 = b_i$, set $m_{d_i} = 1$, shift by d_i , and enter B_i .
2. At B_i , if $g_0 \neq P$ and $m_1 = 0$, shift right in B_i .
3. At B_i , if $g_0 = P, g_1 = E$, write $g_0 = i, g_1 = P$, do not shift, and enter C_{q_i} .
4. At C_q , if $m_0 = 0, g_0 \neq P$, shift left in C_q .
5. At C_q , if $m_0 = 1$, erase the mark and enter A_q without shifting.

For an output in B_i , the current mark distinguishes the first row from the right scan, whose destination is guarded unmarked. In the first case i recovers the overwritten instruction and the guard recovers the old mark. For an output in C_q , a pointer at index one identifies the append and its record at zero identifies i ; a left-scan output has no pointer at index one, by its departure guard. An output in A_q restores the erased mark. This proves injectivity on the full partial domain.

Initially put the pointer at two, with no marks. After n main steps it is at $J = n + 2$, and the new head $h_{n+1} \leq n + 1 < J$. The first rule marks that new head. Scanning, appending, and returning there take exactly $2(J - h_{n+1}) + 3$ rules. Every guard holds; the records occupy $2, \dots, n + 1$ at a main checkpoint. A fresh idle nonhalt start handles an originally halted input when the initial particle is placed outside the observation set.

The three-cell guard. Use the erased-return-mark recorder in Section 5.2, with an idle nonhalt start, odd digits in base $B = 2m + 1$, and N modes. Set $\ell = 1/(100(N + 1)B^6)$. For rational initial codes L_*, R_* , place the nonhalt squares, starting with the initial mode at $j = 0$, at

$$(1/4 - \ell L_* + 2j\ell, 1/4 - \ell R_*),$$

and the halt square at $(3/4, 1/4 - \ell R_*)$, all of side ℓ . The initialized particle is $(1/4, 1/4, 1/4)$. Full triple rectangles have gaps at least ℓB^{-3} ; choose collar $\ell/(10B^3)$. Use the half-period difference in y for their selectors. Set $z_i = 1/2 + i/(4(J + 1))$ for the J branches, with disjoint small height supports. The linear profiles may be

$$(s - c)\chi_{[1/8, 3/8], 1/32}(s) - (s + 1/2 - c)\chi_{[1/8, 3/8], 1/32}(s + 1/2),$$

interpreted through their local periodic charts. They have zero mean and equal $s - c$ in the working strip. Use the seven phases with (2). Their excursion is less than $(1 + B)^4 \ell \leq 16B^4 \ell < 0.02$. Nonhalt source and target squares lie in $0.24 < x < 0.27$, and their full period paths lie in $0.23 < x < 0.28$. Halt endpoints lie in $[3/4, 0.76)$. Consequently the fixed event $2/3 < x < 5/6$ is exactly halting, at all real times.

Marking after a separate work step. Use history alphabet $\{E, F\} \sqcup \{i\}$ for this and the next two recorders. A different six-rule map first writes and moves without testing a mark: at R_q , write $a_0 = b_i$, shift d_i , and enter P_i ; at P_i , require $m_0 = 0$, set $m_0 = 1$, shift right into G_i . In G_i , require $m_0 = 0$ and either scan right when $g_0 \neq F$, or, when $g_0 = F, g_1 = E$, write $g_0 = i, g_1 = F$, shift left into L_{q_i} . In L_q , scan left when $m_0 = 0, g_1 \neq F$; when $m_0 = 1$, erase it and enter R_q without moving. The outputs in P_i retain the overwritten instruction. At G_i , the mark at index -1 distinguishes entry from scan. At L_q , the pointer at index two distinguishes append from scan, and the record at one recovers i . At R_q the erased mark is restored. Hence this is another injective partial map. With frontier $J = n + 2$ its main cycle has $2(J - h_{n+1}) + 2$ rules. It is not identified with the preceding map: the first rule has different guards and the append moves.

The two-cell append. For the separate-work-step map of Section 5.2, use even digits in base $B = 2m + 1$ and the shorter containing interval

$$I(v) = [C(v), C(v) + B^{-|v|}C_*], \quad C_* = 2(m - 1)/(B - 1) < 1.$$

The proof of Lemma 3.1 applies with tail interval $[0, C_*]$; incompatible prefixes have gap at least $(2 - C_*)B^{-j}$ at their first differing position j . Put $\rho = 1/(100(N + 1)(B + 1)^4)$, $Y = 1/2 - \rho R_*$, $X_h = 3/4$, and $X_s = 1/4 - \rho L_* + 2j(s)\rho$ for nonhalt states, starting with the fresh idle start at $j = 0$. At coding height $1/4$, the label is $(1/4, 1/2, 1/4)$. Triple branches have gaps greater than ρB^{-3} ; choose collar $\rho/(10B^3)$. Use half-period differences in y and in height, heights $1/2 + i/(8(J + 1))$ with collar $1/(32(J + 1))$, and (4). The excursion is at most $(B + 1)^4\rho < 1/100$. Nonhalt paths satisfy $x < 1/4 + 1/50 + 1/100 < 2/3$; halt endpoints are in $[3/4, 3/4 + \rho]$. Thus the event is $2/3 < x < 5/6$. The unsloved force is periodic from zero. A separate fourth-root choice $g(t) = (1 + t)^{1/4} - 1$ has the same event and satisfies $\|D_x^\alpha f(t)\|_\infty = O_\alpha((1 + t)^{-3/4})$, with all mixed derivatives bounded, by Proposition 6.2. In particular every spatial derivative of the force is square integrable in time in spatial supremum norm. The periodic and slowed forces are separate prescriptions.

A persistent head mark. For a third map, start with a unique mark at the machine head and retain it at every main visit. Use controls C_q, D_i, R_i, L_q , with a fresh idle start q_b which is no instruction target; omit L_{q_b} . At nonhalt C_q require $m_0 = 1$ and, if $d_i \neq 0$, $m_{d_i} = 0$; write b_i , move the mark to d_i , shift d_i , and enter D_i . At D_i require $m_0 = 1$, shift right into R_i . At R_i require $m_0 = 0$; scan right if $g_0 \neq F$, or append i, F over F, E at indices $0, 1$ and shift left into L_{q_i} . At L_q scan left when $m_0 = 0, g_1 \neq F$; at $m_0 = 1$ enter C_q without editing or moving. The inverse tests are respectively the instruction i , the mark at -1 , the pointer at two with its record at one, and the unchanged marked tape. Thus they apply without assuming unique markers globally. No rule enters C_{q_b} . With the initial history end at two, the main-cycle count is $2(J - h_{n+1}) + 2$. The mark remains at the new machine head throughout logging, and all initialized guards hold.

5.3 A remote center and positive thickness

The persistent-head recorder of Section 5.2 has an initial mode with no incoming rule. Place its mode squares first in unscaled coordinates: the first origin is $2j$ for a main halt and $-2j$ for every other mode, with distinct positive indices j ; the second origin and coding height are zero. Use odd digits in base B and full triples. The rectangle gaps are at least B^{-3} , and half-widths are at most $1/2$. Let N be the number of ordinary branches plus one, let O_0 be the largest absolute mode origin, and set

$$K = (1 + B^2)^4, \quad R = 10N + O_0 + K + 10, \quad L = 64R.$$

The loading branch is the translation of the square of half-width $1/(4B)$ centered at $(-4R, 0)$ to the congruent square centered at the rational initial code. Odd-digit codes have coordinate margins at least $1/(B-1)$, so its target fits inside the mode square. That square has no incoming branch. Both augmented rectangle families are consequently separated.

Choose heights $Z_i = 10i$, source and target collars $1/(4B^3)$, and height cutoffs g_i equal to one on $[Z_i - 1, Z_i + 1]$, supported within a further unit collar. The common center is $c^\circ = (-2R, 0)$. Use eight motions: lift by Z_i , translate the center to c° , perform (3) there, translate to the target center, and lower by Z_i . The central linear profiles are

$$(X + 2R)\chi_{[-K, K], 1}(X + 2R)g_i(Z), \quad Y\chi_{[-K, K], 1}(Y)g_i(Z).$$

All scalar profiles have compact support in $(-L/8, L/8)$ in their dependent coordinates. For a scalar profile a depending on d coordinates define

$$a^\#(v) = \sum_{m \in \mathbb{Z}^d} (a(v - Lm) - a(v - Lm - (L/2)e_1)). \quad (9)$$

This is L -periodic and has zero mean. In the central cube it agrees with a , since every noncentral or negatively translated copy is disjoint there. Replace every scalar profile by this correction.

For every starting height $z \in [-1, 1]$, lifting puts the point in $[Z_i - 1, Z_i + 1]$. The four-shear offset bound is $K/2$, with partial coefficients included, and the two translations keep offsets at most $1/2$. All paths lie in $(-8R, 8R)^3$; every controlling profile remains on its intended plateau and all correction terms vanish. The time-one map is therefore the specified reciprocal affine map on the entire $P_i \times [-1, 1]$, with the height unchanged. This is a neighborhood statement, not merely an assertion on the coding sheet.

Let W denote the resulting L -periodic spatial field and $a = (1/2, 1/2, 1/2)$. At the prescribed unit-torus viscosity ν , define

$$U(t, x) = L^{-1}W(t, L(x - a)), \quad f(t, x) = (L^{-1}W_t - \nu L\Delta W)(t, L(x - a)).$$

The factor of the Laplacian is L . These formulas directly give zero pressure, mean-zero solenoidal forcing, and the same time period. The loading center becomes the fixed label $(7/16, 1/2, 1/2)$. During a nonhalting transition the endpoint centers are at most -1 in first coordinate. Translation paths have $X \leq -1 + 1/2 < 0$; central shear paths have $X \leq -2R + K < 0$. Loading obeys the same bound, and no path crosses the chart seam. A halt endpoint has positive first coordinate. Thus the fixed event on the unit torus is $1/2 < x < 1$, and the loader is part of the period repeated from zero. The locally finite sum (9) is effectively evaluated from known supports and a bounded coordinate enclosure.

5.4 Moving the head and retaining the origin

The persistent map admits a distinct five-rule form. Normalize to one halt state and a different, fresh nonhalting start. Use a return control B_q for every q , including that start; this version does not impose the preceding no-incoming-start restriction. Add an origin bit, fixed at absolute zero and never edited, and combine the first two motions: at M_q , perform the same guarded write and mark move but shift by $d_i + 1$ into O_i . The O_i scan and append and the B_q return and exit are the preceding R_i, L_q rules. An O_i output from the main rule has mark one at index -1 ; a scan output has mark zero there. The append/return distinction is still the pointer at two. This proves the full inverse, including the possible shifts zero and two. At a main checkpoint the history end J satisfies $J \geq h + 2$. After the work move to $h' = h + d_i$, the working cursor is $h' + 1 \leq J$. There

are $J - 1 - h'$ outward and the same number of backward scan steps, so one machine transition uses $2(J - h') + 1$ rules. The origin flag recovers absolute indexing independently of the history. Resolving on $[-2, 1]$ gives prefix lengths 2, 2 and target lengths $2 + d, 2 - d$ for each actual working shift $d \in \{-1, 0, 1, 2\}$; an empty target prefix when $d = 2$ is allowed in Lemma 3.1.

The preceding three guarded recorders use complete triples on $[-1, 1]$, with target prefix lengths $1 + d, 2 - d$. Thus all four maps provide the full closed rectangle interface. The initialized marker invariant proves halting equivalence; the separate inverse proofs prove geometric separation.

A head marker and a separate origin flag. Use the five-rule map just proved. Set $\ell = 1/(100(N + 1))$, put its start origin at $(1/4, 1/4) - \ell(L_*, R_*)$, its halt origin at $(3/4, 1/4)$, and its other origins at $(1/4 + (3 + 2j)\ell, 1/4)$. Nonhalt coordinates are at most $27/100$. Resolve the four-cell window and use odd digits in base $2m + 1$. The possible scale includes B^{-2} , but both endpoint side lengths are at most ℓ . Derivative selectors and (2) therefore have excursion $3\ell/2$ by the endpoint-size estimate. Every path lies in the linear chart and a nonhalting one satisfies $x \leq 27/100 + 3\ell/2 < 1/2$. Halt endpoints lie in $[3/4, 3/4 + \ell]$. This gives the label $(1/4, 1/4, 1/4)$ and event $1/2 < x < 7/8$.

In each of these cases the rectangle identity proves the code evolution by induction at integer times. The recorder's positive finite cycle counts prove that every machine step is completed. The displayed bounds exclude the observer during every phase of every transition whose source and target are nonterminal. The final transition into a terminal code is allowed to enter the observer before its endpoint. These facts give the claimed equivalence at every real time, rather than merely at the sampling times.

6 Loading once, changing the clock, and choosing the chart

6.1 Onto clocks and the exact derivative estimates

Slow forcing must still complete every finite logical step. The relevant condition is that the new clock maps the nonnegative half-line onto itself. The following elementary calculation applies to our transverse pulse templates, including their finite loading intervals.

Lemma 6.1 (Onto clocks). *Let $W(s, x)$ be a smooth divergence-free field on a flat torus, with zero mean, $W(0) = 0$, bounded mixed derivatives and $(W \cdot \nabla)W = 0$. Let $g : [0, \infty) \rightarrow [0, \infty)$ be smooth, nondecreasing and onto, with $g(0) = 0$. Then*

$$\begin{aligned} u(t, x) &= g'(t)W(g(t), x), \\ f(t, x) &= g''(t)W(g(t), x) + (g'(t))^2 W_s(g(t), x) - \nu g'(t) \Delta W(g(t), x) \end{aligned}$$

give a mean-zero solenoidal force and a solution with zero pressure and zero initial velocity. Its flow is $X_u(t, a) = X_W(g(t), a)$, so every all-time reachability event is unchanged. For $g(t) = \log(1 + t)$ and every k, α ,

$$\|\partial_t^k D_x^\alpha u(t)\|_\infty + \|\partial_t^k D_x^\alpha f(t)\|_\infty \leq C_{k,\alpha}(1 + t)^{-1-k}.$$

All these derivatives are square integrable in time in supremum norm, and in space-time.

Proof. The chain rule gives $f = u_t - \nu \Delta u$. Divergence, mean zero, and vanishing self-advection persist under multiplication by the scalar $g'(t)$ and time composition. The same chain rule proves the trajectory identity by ODE uniqueness. Onto ensures that every finite logical time has a finite physical preimage, proving both implications for the event. The energy comparison in Lemma 2.1 applies to this explicit solution on every finite interval, without a periodicity assumption.

For the logarithmic choice put $a = (1 + t)^{-1}$ and $s = \log(1 + t)$. The velocity is $aW(s)$ and the force is $-\nu a \Delta W(s) + a^2(W_s - W)(s)$. For each positive integer r ,

$$\partial_t[a^r A(s, x)] = a^{r+1}(\partial_s - r)A(s, x).$$

Repeated differentiation and the bounded template derivatives give the stated estimate. Its square is integrable, and the torus has finite volume. These are estimates for the precomputed force formula and require no simulation of the instructions. $\square$

Proposition 6.2 (Cube-root and fourth-root clocks). *Under the hypotheses on W in Lemma 6.1, choose $g(t) = (1 + t)^q - 1$, with $q = 1/3$ or $1/4$, and put $\beta = 1 - q$. The resulting force and velocity have bounded mixed derivatives and*

$$\|\partial_t^k D_x^\alpha u(t)\|_\infty + \|\partial_t^k D_x^\alpha f(t)\|_\infty \leq C_{k,\alpha}(1 + t)^{-\beta}$$

for every fixed k, α . Each such derivative is square integrable in time in spatial supremum norm.

Proof. Every positive derivative $g^{(j)}$ is bounded and is $O_j((1 + t)^{-\beta})$. Every differentiated term of $g'W(g)$ contains at least one such factor and a bounded derivative of W . One factor supplies the decay and the remaining factors are bounded. The identity $f = u_t - \nu \Delta u$ proves the same estimate for the force. Finally $2\beta > 1$, and each clock is increasing and onto. $\square$

The force obtained after slowing is a different prescription from the unsloved periodic force. In particular these conclusions do not combine decay with physical period one. They also do not imply decay faster than every power of time for an arbitrary repeated processor: a speed bounded by $C(1 + t)^{-2}$ has only finite accumulated logical time. The same obstruction appears in the viscous example of Cardona–Miranda–Peralta-Salas–Presas [3, Section 6B]: their exponentially damped velocity traverses only a finite interval of the underlying stationary trajectory. The onto condition above ensures that slowing preserves every finite stage of the computation.

6.2 A separate loader and changes of clock

The following constructions keep the repeated processor independent of the input. A horizontal transverse pulse depending only on height first sends a fixed particle to the rational initial code. Its height profile has mean zero and value one at the coding height. The loader occupies $[0, 1]$ and the processor starts afterward.

For the delayed-move table (8) with frontier at two, let $\ell = 1/(32(N + 1)(B + 2)^4)$ and place mode s at $(\xi_s, 1/4 + 2\ell i(s))$, where $1 \leq i(s) \leq N$ and $\xi_s = 1/4$ or $5/8$ according to halting status. Use derivative rectangle masks, height selectors $p_i = \zeta_i$ and $q_i = \partial_z((z - z_i)\zeta_i)$, and mean-zero linear profiles $\partial_s(\frac{1}{2}(s - c)^2\chi(s))$, with $z_i = 1/2 + i/(4(J + 1))$. Here p_i need not have zero mean: the linear profiles provide it in the four shear phases. The seven phases (4) have excursion at most $1/(64(N + 1))$. Load from $(1/4, 1/4, 1/4)$, using the same derivative height recipe at height $1/4$. In a nonhalting run both the loader and every processor path have $x < 1/2$; terminal codes lie near $5/8$. The observer is $1/2 < x < 7/8$. The fourth-root clock gives

$$\|\partial_t^k D_x^\alpha u(t)\|_\infty + \|\partial_t^k D_x^\alpha f(t)\|_\infty = O_{k,\alpha}((1 + t)^{-3/4}),$$

for every fixed k, α , by Proposition 6.2. The codes occur at physical times $(2 + j)^4 - 1$.

6.3 The move-first recording input

For an instruction $r = (q, a)$ with data (q_r, b_r, d_r) , the recorder of [10, Lemma 2.1] is written here with its frontier symbol P renamed F and its placement controls F_q renamed P_q ; all other controls and guards are unchanged. It uses letters (a, h, m) , $h \in \{E, F\} \sqcup \{[r]\}$ and $m \in \{0, 1\}$, with modes S_q, A_r, R_r, P_q, L_q . Here is its complete table; every occurrence of h requires $h \neq F$:

S_q	$(a, h, 0)$	$(b_r, h, 0)$	d_r	A_r
A_r	$(c, h, 0)$	$(c, h, 1)$	1	R_r
R_r	$(c, h, 0)$	$(c, h, 0)$	1	R_r
R_r	$(c, F, 0)$	$(c, [r], 0)$	1	P_{q_r}
P_q	$(c, E, 0)$	$(c, F, 0)$	-1	L_q
L_q	$(c, h, 0)$	$(c, h, 0)$	-1	L_q
L_q	$(c, h, 1)$	$(c, h, 0)$	0	S_q .

There are no first-row instructions for halting q . The cited lemma proves that the target mode determines the incoming move, and the target mode with the written complete letter determines the unique predecessor. Initialize at the original machine head with frontier at any fixed $r_0 \geq 2$, empty history elsewhere, and zero marks everywhere. After n machine steps the frontier is $r_0 + n$; the work state, tape and head are recovered at the S -visits. If the next machine head is h' , its next checkpoint takes exactly $2(r_0 + n - h') + 4$ local transitions. In particular these counts are finite and positive. Halting and indefinitely defined nonhalting runs are preserved. We use $r_0 = 2$ below.

Section 5.1 of [10] proves two extensions on the whole partial domain: toggle the current mark only in ready states to obtain the marker-preserving version, or copy an arbitrary passive track at every write. The latter permits a permanent origin flag. We specify the two changed rows again when the whole-box application uses the former. Every other recorder in this paper has its own complete inverse proof.

The move-first processor with a separate loader. For the seven-row move-first recorder just recalled, retain all its original halt labels and put $h = 1/(100(N + 1)(1 + B)^4)$, with mode origins $(a_s, 1/4 + 2jh)$, where $a_s = 1/4$ for nonhalting modes and $a_s = 3/4$ for halting modes. Use collar $h/(4B^2)$, rectangle selectors corrected by a half-period shift in y , and height selectors corrected by a half-period shift in z . Choose processing heights $1/2 + i/(4(J + 1))$ and support radii $1/(16(J + 1))$. The seven phases (4) have excursion at most $1/100$. Load from $(1/4, 1/4, 1/4)$ with a half-period-corrected selector of radius $1/32$ about height $1/4$. The nonhalting paths remain in $[1/4 - 1/100, 1/4 + h + 1/100]$; terminal codes are near $3/4$. The event is $2/3 < x < 9/10$. One choice is the unslowed force, periodic after loading. Another is $g(t) = \log(1 + t)$, giving $\|f(t)\|_\infty = O((1 + t)^{-1})$, square-integrable forcing, and bounded mixed derivatives, by Lemma 6.1.

Both loaders move along a straight segment. If the initial machine is already halted that segment is permitted to enter the observer; in a nonhalting computation its endpoint is in the nonhalting band and the whole segment avoids it. This explicitly includes the initial case in the continuous-time argument.

6.4 Eight phases about the center of the chart

Use the seven-row move-first recorder of Section 6.3 with one halt label. For its odd-digit alphabet let the base be $D = 2m + 1$. If there are n nonterminal modes, set $\lambda = 1/(16(n + 1))$, use common second origin $3/8$, nonterminal first origins $1/4 + 2j\lambda$ ($0 \leq j < n$), and halt origin $5/8$. Refine every single-cell rule by its left letter, including right and stay moves. Its source then has side lengths

λ/D in both directions. Formula (7) gives target rectangles; the right-move target is $T_b(I_c) \times [0, 1]$, the stay target is $I_c \times I_b$, and the left-move target is $[0, 1] \times T_c(I_b)$. The target separation follows from the incoming rule and then the extra left letter. Both families have gaps at least λ/D^2 .

Let branch i have centers c_i, c'_i and reciprocal factor μ_i . Number the J branches by $j_i = 0, \dots, J-1$, put $B_0 = \max(2, J+1)$, and use coding height zero. Source and target cutoffs with collars $\lambda/(8D^2)$ have second-coordinate supports in $(1/4, 1/2)$. Therefore

$$w^\pm(x, y) = \sum_i j_i (\chi_i^\pm(x, y) - \chi_i^\pm(x, y - 1/2))$$

have zero mean and value j_i on the appropriate rectangle. For every $j < B_0$, choose a periodic cutoff P_j equal to one within radius $1/(8B_0)$ of j/B_0 , supported within radius $1/(4B_0)$, and set $G_j(z) = P_j(z) - P_j(z - 1/(2B_0))$. Then G_j has zero mean and $G_j(k/B_0) = \delta_{jk}$, including the height at the coordinate seam.

Put $c_* = (1/2, 1/2)$ and let $h(s) = (s - 1/2)\chi(s)$ be a periodic profile with $\chi = 1$ on $[3/8, 5/8]$ and support in $[1/4, 3/4]$. At height j_i/B_0 , perform these eight motions: lift by j_i/B_0 , translate by $c_* - c_i$, apply

$$X(1), \quad Y(\mu_i^{-1} - 1), \quad X(-\mu_i), \quad Y(-(\mu_i^{-1} - 1)/\mu_i),$$

translate by $c'_i - c_*$, and lower by j_i/B_0 . The lift and lower use $\pm B_0^{-1}w^\pm e_z$; translations use $G_{j_i}(z)$; each central shear uses the same selector and the appropriate profile h . Thus all fields have zero mean and zero self-advection.

To verify the plateaus, let the initial centered offsets be (ξ, η) , with $|\xi|, |\eta| \leq \lambda/(2D)$. The successive endpoints are

$$(\xi + \eta, \eta), \quad (\xi + \eta, (\mu^{-1} - 1)\xi + \mu^{-1}\eta), \quad (\mu\xi, (\mu^{-1} - 1)\xi + \mu^{-1}\eta), \quad (\mu\xi, \mu^{-1}\eta).$$

Since $\mu, \mu^{-1} \leq D$, all second deviations are at most λ and all first deviations at most $\lambda/2$; partial shears interpolate between these endpoints. The profiles remain linear, and the complete endpoint is the prescribed target point at height zero.

Load the fixed particle $(1/2, 1/2, 0)$ to the rational initial code by a horizontal pulse with height selector G_0 . In a nonhalting run all endpoint centers have first coordinate in $[1/4, 3/8]$. During the centered shears $|x - 1/2| \leq \lambda/2$; translations and loading have the same upper bound. Hence $x \leq 1/2 + \lambda/2 < 9/16$ throughout. A halt code lies in $[5/8, 5/8 + \lambda] \subset (9/16, 3/4)$. The fixed event is therefore $9/16 < x < 3/4$. The cube-root clock $g(t) = (1 + t)^{1/3} - 1$ gives the samples $(2 + j)^3 - 1$ and, by Proposition 6.2, for every k, α ,

$$\|\partial_t^k D_x^\alpha u(t)\|_\infty + \|\partial_t^k D_x^\alpha f(t)\|_\infty = O_{k,\alpha}((1 + t)^{-2/3}).$$

Every such derivative is square integrable in time in supremum norm.

6.5 Keeping the physical scale of a side-ten construction

The seven-row move-first table also admits a useful larger chart. Normalize the original machine to one halt label before constructing this table, and let Γ be its augmented tape alphabet. Add a copied origin bit at absolute cell zero, put the frontier at two, and retain the table's guards on every other track. The incoming displacement and written-letter inverse are unchanged. If there are m modes, set $r = 1/(4m)$, put nonhalt origins at $(2 + 2rj, 2)$ and the halt origin at $(6, 2)$, and use coding height two in $\mathbb{T}_{10}^3$. Odd digits in base $K = 2|\Gamma| + 1$ and (7) give gaps at least r/K^2 . Use rectangle collars $d = r/(4K^2)$ and subtract their copies shifted by four in y . Use processing heights

$Z_i = 4 + i/(J + 1)$ with collar $1/(4(J + 1))$, subtracting copies shifted by three in z . The linear profile around a center c is $(s - c)\chi_{[c-1/2, c+1/2], 1/2}(s)$.

Apply (2) between lift, translation, and lower. The source half-widths α, β satisfy

$$\alpha, \beta, a\alpha, \beta/a \leq r/2.$$

Its second intermediate horizontal coordinate is $a\xi + (a^{-1} - 1)\eta$; its modulus is at most r , since $|a^{-1} - 1|\beta = |\beta/a - \beta| \leq r/2$. All other intermediate deviations are at most r as well. Hence every profile is linear on the required paths. A mean-zero height pulse at height two loads $(2, 2, 2)$ to the rational initial code. A nonhalting path stays in $0 < x < 5$, while halt codes have $6 \leq x \leq 6 + r$. The fixed event is $5 < x < 8$. This is a side-ten statement at the given physical viscosity ν .

For completeness, a general scale change from side ten with viscosity ν_{10} is

$$\begin{aligned} v(s, y) &= \frac{T}{10}U(Ts, 10y), & p(s, y) &= \frac{T^2}{100}P(Ts, 10y), \\ f(s, y) &= \frac{T^2}{10}F(Ts, 10y), & \nu_1 &= \frac{T}{100}\nu_{10}. \end{aligned} \tag{10}$$

Every term of the equation has factor $T^2/10$. With $T = 1$ the label becomes $(1/5, 1/5, 1/5)$ and the event $1/2 < x < 4/5$; to obtain a prescribed unit-torus viscosity ν one must start with $\nu_{10} = 100\nu$. Keeping the viscosity unchanged instead requires $T = 100$ and changes the time period. These are coordinate transformations with their stated physical parameters.

7 Transferring a whole box of positive thickness

7.1 Sequential parking of whole boxes

We next protect every point in a box of fixed positive thickness. Let $J_0 = [-1/2, 1/2]$. Suppose closed rational rectangles P_i, Q_i are separately pairwise disjoint and have positive side lengths at most one. Their prescribed center-to-center affine maps F_i satisfy $F_i(P_i) = Q_i$ and have reciprocal factors $\text{diag}(a_i, a_i^{-1})$, $a_i > 0$ rational. The box map fixes the third coordinate on $P_i \times J_0$. There is no requirement that a source miss another target.

Lemma 7.1 (Parking before delivery). *The specified maps have an effective smooth realization on their whole source boxes by successive transverse shear pulses. The velocity is zero near the endpoints of its unit time interval, and all derivatives are bounded. Every scalar profile is compactly supported in the coordinates on which it depends. If a source and its target lie entirely in $x < 0$, that whole moving box stays in $x < 0$ throughout the realization.*

Proof. For an empty list use zero. Otherwise let $x_{\min}$ be the least first coordinate of any source or target. Set

$$g_i = (\min(0, x_{\min}) - 5 - 3i, 0), \quad G_i = g_i + [-1/2, 1/2]^2.$$

The parking squares are separated from each other and from every source and target. Let d_0 be the minimum of one and all positive within-list sup-norm distances for sources, targets, and parking squares, together with all distances between a parking square and a source or target. Omit empty lists in this minimum and use collar $\varepsilon = d_0/4$. Rectangle cutoffs equal one on neighborhoods and vanish outside these collars. Put $Z = 4$ and $J_Z = Z + J_0$.

First evacuate every source in turn: lift it by Z , translate its center to g_i at height J_Z , and lower it. Only after every source has been vacated, process the parked boxes one at a time: lift,

apply the four shears (2) about g_i , translate to its target center, and lower. For l boxes these passes use $3l + 7l = 10l$ slots, placed inside $[1/4, 3/4]$. Run every profile with a smooth nonnegative unit-integral pulse in its slot.

A vertical motion over its footprint C and a horizontal translation use the respective profiles

$$\pm Z\chi_C(x, y)e_z, \quad (b_1, b_2, 0)\chi_{J_Z}(z).$$

A shear adding $c(y - g_{i,2})$ to x uses

$$c(y - g_{i,2})\chi_{[g_{i,2}-2, g_{i,2}+2]}(y)\chi_{J_Z}(z)e_x,$$

and analogously for the other direction. These are transverse fields. Write the initial offset as (ξ, η) . Source and target sizes give $|\xi|, |\eta|, |a_i\xi|, |\eta/a_i| \leq 1/2$. The four endpoints are

$$(\xi, \eta - a_i\xi), \quad (a_i\xi + (a_i^{-1} - 1)\eta, \eta - a_i\xi), \quad (a_i\xi + (a_i^{-1} - 1)\eta, \eta/a_i), \quad (a_i\xi, \eta/a_i).$$

Their second deviations are at most one and first deviations at most $3/2$. A partial shear follows the segment between its endpoints. All controlling deviations therefore remain in $[-2, 2]$, so the cutoff computation is valid on every point of the box.

During evacuation every other material box is an unlifted source or an already parked box; during delivery it is a parked box or a completed target. Footprint separation makes every vertical pulse vanish on these other boxes. Horizontal pulses are supported near J_Z and vanish on J_0 , since $\varepsilon \leq 1/4$. Thus all the stated moving and stationary paths solve the same smooth ODE. Uniqueness proves the whole-box claim by induction over slots. Finally $g_{i,1} \leq -8$ and centered shear excursions are at most two. When both endpoints are negative, both translations interpolate between negative coordinates; vertical motion does not affect them. This proves the sign assertion. Compact transverse cutoffs give all derivative bounds. $\square$

Normalize the original machine to one halt label before constructing its recorder. To apply the lemma, take the marker-preserving version of that seven-row recorder. Explicitly, change its first ready rule to read mark one and write mark zero, and its last return rule to write mark one; initialize one mark at the head. All intermediate rows are unchanged. The written symbol still distinguishes every incoming case; at a ready checkpoint the unique mark is retained rather than erased. The first row removes it, the next marks the new head, and the return retains it. This proves the same full-domain inverse and initialized simulation directly. Use odd digits and (7), and place its unit mode squares at $(\alpha_s, 0)$, where $\alpha_h = 1$ and other $\alpha_s = -2j$, $j \geq 1$. Thicken every branch by J_0 . The initial code y_{in} is rational and has height zero. Nonhalting boxes lie wholly in $x < 0$; halting codes lie in $1 < x < 2$.

Apply Lemma 7.1 to the stepping list, and separately to the single loading translation from the cube $[-1/2, 1/2]^3$ to the congruent cube centered at y_{in} . If there are N modes and l stepping branches, take $R_* = 100 + 2N + 4\max(1, l)$. All paths and bounded transverse supports lie strictly in $(-R_*, R_*)$ in each relevant coordinate: the original rectangles lie in $[-2N - 1, 3] \times [-1, 2]$; parking adds at most $5 + 3\max(1, l)$ to the negative first extent; shearing adds at most two and a collar at most $1/4$; heights are below $4 + 1/2 + 1/4$. This also covers the loading cube.

Scale by $\lambda = (8R_*)^{-1}$ into the unit torus centered at zero. For each pulse with transverse index set I , form its scaled periodic field $F(x) = \lambda v(x_I/\lambda)$ and replace it by $F(x) - F(x + e_j/2)$, where $j = \min I$. Its transverse support is inside $(-1/8, 1/8)^I$; the translated term vanishes on every tracked path. The corrected pulse has zero mean and still has zero divergence and self-advection. Concatenate the loading interval and the repeated stepping interval and prescribe $f = U_t - \nu\Delta U$.

Lemma 2.1 gives the resulting smooth solution, pressure zero, and uniqueness in the stated classical class.

Write κ for this configuration code, $\mathcal{T}$ for the recorder transition, and c_0 for its initialized configuration. The fixed label is the torus origin, and the event is the positive half-circle $0 < x < 1/2$. At time $1 + j$ its code is $[\lambda\kappa(\mathcal{T}^j c_0)]$. A halt gives a point in that event, including a terminal initial configuration. For a nonhalting computation the sign part of the lemma excludes the event during stepping. During loading, the origin is the center of its cube: it first has coordinate zero, then moves to a negative parking center, is fixed relative to that center by every shear, and translates to the negative initial code. Its first coordinate is nonpositive throughout. No scaled path leaves $(-1/8, 1/8)^3$, so reduction modulo one introduces no spurious visit to the positive half-circle.

The unsloped force is periodic after loading, with its repeating part depending only on the machine table. A separate logarithmic choice uses Lemma 6.1; every mixed derivative of force is bounded and square integrable on space–time. It reaches code j at $t = \exp(1 + j) - 1$. These claims apply to the displayed finite pulse prescription, whose construction never executes the simulated machine.

8 When a rule changes planar area

Reciprocal scaling is appropriate for a head move between two tape halves: one half is shortened exactly as much as the other is lengthened. A prefix rule that also inserts a history symbol need not preserve planar area. It can still act on a coding sheet of an incompressible flow, provided the normal direction compensates for the change of area. The contraction, transport, and expansion organization has a geometric antecedent in Cardona–Miranda–Peralta–Salas–Presas [3, Proposition 5.1]; here explicit transverse shears supply the prescribed sheet maps and the normal compensation. We first prove Theorem 1.2 for arbitrary rectangle data by evacuating the source sheets into storage and processing them one at a time. The proof also controls their low-height positions, which will give a detector for a separator-stack recorder. We then give a different recorder and a fifteen-phase construction that processes all its branches simultaneously at separate heights. In both constructions the normal action is computed explicitly.

8.1 Sequential realization of arbitrary sheet maps

Take the data of Theorem 1.2: separately separated closed rational rectangles $P_i, Q_i \subset [2, 3]^2$, with positive side lengths, and their prescribed positive diagonal affine maps $F_i : P_i \rightarrow Q_i$. We realize every map on $P_i \times \{2\}$ in one period. For an empty family use the zero field; henceforth let $1 \leq i \leq N$ with $N > 0$.

Write v_i, v'_i for the centers of P_i, Q_i and h_i, h'_i for their half-width vectors. All these half-widths are at most $1/2$. Take one tenth of the minimum of one and the within-source and within-target sup-norm gaps as $d > 0$, omitting an empty list of gaps. Thus $d \leq 1/10$. With an even cutoff $\chi_h = 1$ on $[-h - d/2, h + d/2]$ and zero outside $[-h - d, h + d]$, define side-ten periodic profiles

$$C_{h,c}(s) = \sum_{k \in \mathbb{Z}} \chi_h(s - c - 10k), \quad D_{h,c}(s) = \sum_{k \in \mathbb{Z}} (s - c - 10k) \chi_h(s - c - 10k).$$

The D profile has mean zero by oddness. Vertical transport over a footprint centered at v with half-widths h uses

$$(z^+ - z^-) e_z C_{h_1, v_1}(x) (C_{h_2, v_2}(y) - C_{h_2, v_2+4}(y)).$$

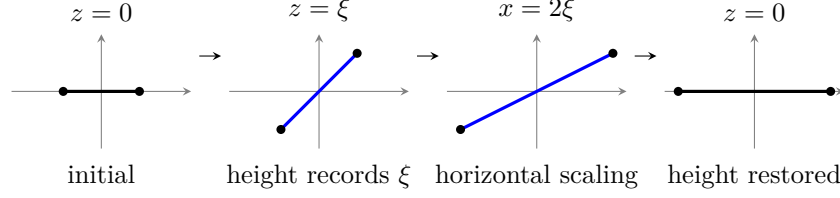


Figure 1: A slice of the sheet-scaling triple, in local coordinates with vertical coefficient one and horizontal factor $\lambda = 2$. A vertical shear records the initial horizontal offset in height; the horizontal shear doubles that offset; the last vertical shear returns the sheet to height zero. Nearby normal offsets are contracted by $1/2$, as the determinant-one formula in the text shows. Each depicted segment is the image of the entire initial segment.

Horizontal transport from center v^- to center v^+ at height z uses

$$((v^+ - v^-), 0)(C_{0,z}(x_3) - C_{0,z+3}(x_3)).$$

These have zero mean and the indicated exact translations on plateaus.

Changing a sheet coordinate. Use coding height two, transport height five, storage shift $(4, 0)$, and processing center $v_* = (5, 6)$. At this center, first-coordinate scaling uses $H_1 = D_{1/2,5}(x)C_{1/2,6}(y)$, second-coordinate scaling uses $H_2 = C_{1/2,5}(x)D_{1/2,6}(y)$, and put $B(z) = D_{1/2,5}(z)$. For $j = 1, 2$ and $\lambda = h'_{ij}/h_{ij}$ use three phases

$$e_z H_j, \quad e_j(\lambda - 1)B(z), \quad -e_z H_j/\lambda. \quad (11)$$

At the processing center, an offset r_j goes to height $5 + r_j$, then to horizontal offset λr_j , then back to height five. All unscaled and scaled half-widths are at most $1/2$; every intermediate horizontal offset is between these endpoints. Therefore the cutoffs stay linear and heights lie in $[9/2, 11/2]$. For a small height deviation ζ the local two-dimensional action is $(r_j, \zeta) \mapsto (\lambda r_j + (\lambda - 1)\zeta, \zeta/\lambda)$, whose determinant is one. The sheet returns to height five, while a nearby normal displacement is divided by the horizontal expansion factor. Applying the two triples therefore compensates for the full planar area change; Figure 1 shows one coordinate triple.

Evacuating before delivery. For N rectangle maps schedule $13N$ pulses inside $[1/4, 3/4]$. First lift each source sheet to five, shift by $(4, 0)$, and lower it into storage at height two. After all sources have been evacuated, process them in order: lift the stored sheet, translate its center to v_* , perform the six phases (11), translate to its target center, and lower. These passes take three and ten phases per sheet. The maps may send a source onto another source: the preliminary evacuation ensures that every target footprint is free when delivery begins.

Every resting sheet has height two and second coordinate in $[2, 3]$. During evacuation, the other resting sheets are unprocessed sources or stored sheets; during delivery, they are stored sheets or completed targets. The within-family gaps and the separation between the coding and storage squares therefore keep each vertical support off all other resting footprints. The correcting row has second coordinate in the d -collar of $[6, 7]$ and misses the resting sheets. Horizontal transport is supported near heights five and eight. The H_j profiles are supported near second coordinate six and the B profile near height five. Thus all processing pulses fix every resting sheet. All supports in the dependent coordinates and all moving paths are interior to the side-ten chart; periodic copies

create no further intersection. By induction through the slots, the full source sheets have precisely their prescribed affine images. The preliminary evacuation is what permits arbitrary intersections between the source and target families.

Repeat this finite sequence from time zero and apply Lemma 2.1. Every phase is transverse and mean zero, and the rational data and fixed smooth profiles make the construction effective. This proves Theorem 1.2, including its bounded-derivative, zero-pressure, and periodicity conclusions.

A bound for every intermediate position. The same geometry gives a useful observation rule for each moving point. Whenever the height lies in $(3/2, 5/2)$, the first coordinate belongs to

$$\{x^-, x^- + 4, x^+\}, \quad (12)$$

where x^-, x^+ are that point's source and target first coordinates. Horizontal transport takes place at height five and scaling in $[9/2, 11/2]$; in the indicated low band only waiting and vertical motion occur over the source, storage point, or target. This proves (12) throughout the period.

8.2 A separator-stack application and its observer

We now supply rectangle data to the sequential construction. The recorder separates a finite left working word from its retained history. A prefix rule $(s; \alpha, \beta) \rightarrow (s'; \bar{\alpha}, \bar{\beta})$ replaces the two indicated finite stack prefixes and retains both arbitrary infinite tails. Normalize the original machine to one halt label, and let A be its tape alphabet, with blank b_0 . Use the stack alphabet $A \sqcup \{\perp, \diamond\} \sqcup \{\eta_i\}$ and controls R_q, T_i, S_p . At a ready control R_q , the left stack has the form $L = \ell \perp H$, with $\ell \in A^*$ listing the working cells to the left of the head, nearest first, and implicit blanks beyond them. The separator $\perp$ keeps these cells above the retained history H ; the right stack begins with the scanned cell. A temporary symbol $\diamond$ marks the beginning of the updated right working word while T_i transfers the finite left word onto that stack above the marker. The control S_p then restores that word until $\diamond$ is exposed and removed. A new symbol η_i retained below $\perp$ records the instruction case. In the transfer and restore rules, c ranges over A . Give each applicable case of an original instruction $(q, a) \mapsto (p, b, d)$ its own index i , with $p(i) = p$, and use

$$\begin{array}{l|l} d = 0 & (R_q; \emptyset, a) \rightarrow (T_i; \emptyset, \diamond b), \\ d = 1 & (R_q; \emptyset, a) \rightarrow (T_i; b, \diamond), \\ d = -1, c \in A & (R_q; c, a) \rightarrow (T_i; \emptyset, \diamond cb), \\ d = -1, \text{ empty left} & (R_q; \perp, a) \rightarrow (T_i; \perp, \diamond b_0 b). \end{array} \quad (13)$$

Complete the table with

$$\begin{array}{ll} (T_i; c, \emptyset) \rightarrow (T_i; \emptyset, c), & (T_i; \perp, \emptyset) \rightarrow (S_{p(i)}; \perp \eta_i, \emptyset), \\ (S_p; \emptyset, c) \rightarrow (S_p; c, \emptyset), & (S_p; \emptyset, \diamond) \rightarrow (R_p; \emptyset, \emptyset). \end{array} \quad (14)$$

Initialize with $(R_{q_0}, \perp b_0^\infty, w b_0^\infty)$. At a ready state with $L = \ell \perp H$, the primary rule yields $(T_i, \ell' \perp H, \diamond Y')$ with the correct updated tape. The transfer moves ℓ' right in reverse order, inserts η_i below $\perp$, and the restore returns precisely those work symbols until $\diamond$ is exposed and removed. The new ready configuration is $(R_{p(i)}, \ell' \perp \eta_i H, Y')$, after $2|\ell'| + 3$ rules. This is finite and positive, including an empty left word. Its special left-move case exposes an implicit blank correctly. The retained instruction-case records also determine the displacement history if absolute indexing is desired.

The source families are disjoint by tested control and top symbol. Into T_i , the unique primary arrival has right top $\diamond$; loop arrivals have distinct real symbols. Into S_p , transfer arrivals have left

prefix $\perp\eta_i$, while loops have real left top; record indices and symbols distinguish their respective cases. There is one incoming rule into each R_p . This proves disjoint image cylinders on arbitrary tails and gives their inverses.

Use a common digit dictionary for controls and stack symbols. If there are m entries other than the halt control, let $K = 8(m + 1)$, give them distinct digits $2, 4, \dots, 2m$, and give the halt control digit $3K/4$. Set $e(v) = \sum_j d(v_j)K^{-j}$ and $I(v) = [e(v), e(v) + K^{-|v|}]$ for finite prefixes. The code is

$$(s, L, Y) \mapsto (2 + e(sL), 2 + e(Y), 2) \in \mathbb{T}_{10}^3.$$

For a rule $(s; \alpha, \beta) \rightarrow (s'; \bar{\alpha}, \bar{\beta})$, its rectangles are

$$P_i = (2 + I(s\alpha)) \times (2 + I(\beta)), \quad Q_i = (2 + I(s'\bar{\alpha})) \times (2 + I(\bar{\beta})).$$

Both families are separated in $[2, 3]^2$. Their affine factors are $K^{|\alpha| - |\bar{\alpha}|}$ and $K^{|\beta| - |\bar{\beta}|}$; the control digit cancels in the first ratio. Nonhalt first coordinates are at most $2 + (2m + 1)/K < 9/4$, while halt codes are between $11/4$ and 3 . The initial code is rational by summing the blank tails.

These rectangles and maps satisfy every hypothesis of the sequential construction in Section 8.1. Load $(2, 2, 2)$ horizontally to the rational initial code during $[0, 1]$, then repeat its schedule. The fixed event is

$$5/2 < x < 7/2, \quad 3/2 < z < 5/2.$$

A halt code is in this set, including an initially halted input after loading. In a nonhalting run the loader has $x < 9/4$. During every later period both endpoint coordinates lie in $[2, 9/4)$; (12) makes the low-height first coordinate either less than $9/4$ or at least six. At other heights the event is excluded by its height condition. Thus there is no false intermediate hit even though an upper horizontal route may cross the observer's projection. Idle collars make each integer-time halt witness persist on an interval.

Direct transverse forcing has zero pressure and is periodic after loading at the specified side-ten viscosity. The logarithmic choice of Lemma 6.1 instead has $\|f(t)\|_\infty = O((1 + t)^{-1})$, bounded mixed derivatives, and $f \in L^2([0, \infty) \times \mathbb{T}_{10}^3)$ with the same event. This is a separate choice of forcing. Under the time-preserving scale change (10), the label is $(1/5, 1/5, 1/5)$ and the event is $1/4 < x < 7/20$, $3/20 < z < 1/4$, with the viscosity transformation stated there.

8.3 Prefix rules with retained instruction tags

The next recorder retains history as tags between the work symbols. Its rectangles will be arranged in a narrow strip so that a simultaneous construction can contract, translate, and expand all branches together. Let Σ be the machine alphabet, with blank β . Replace a real tape symbol by $(a, o) \in \Sigma \times \{0, 1\}$, where o marks the original cell zero and is preserved by every write. Write the enlarged transition as $(q, a) \mapsto (q', b, d)$, with $d \in \{+, 0, -\}$. Use controls $M(q, e)$ for $e \in \{\star, +, 0, -\}$ and $P_+(q), P_-(q)$. For every nonhalting main control $s = M(q, e)$ and real symbol a introduce a distinct tag $\gamma_{s,a}$. The stack alphabet $\mathcal{A}$ is the disjoint union of these tags and the real symbols. At a main control $M(q, e)$, erasing all tags gives the two tape halves: the left stack lists cells to the left of the head, nearest first, and the right stack begins with the scanned cell. The control index e records the preceding move, with $\star$ reserved for initialization. The transfer controls $P_+(q)$ and $P_-(q)$ finish a right or left move by passing intervening history tags until the next real work symbol is available. Thus history occupies space in the stacks without being mistaken for a work cell.

A prefix rule $[s; \ell, r] \rightarrow [s'; \ell', r']$ replaces the two displayed prefixes, retaining arbitrary infinite tails. For each main instruction use

$$[s; \emptyset, a] \longrightarrow \begin{cases} [P_+(q'); b\gamma_{s,a}, \emptyset], & d = +, \\ [P_-(q'); \emptyset, b\gamma_{s,a}], & d = -, \\ [M(q', 0); \emptyset, b\gamma_{s,a}], & d = 0. \end{cases} \quad (15)$$

For every tag g and real symbol c include

$$\begin{aligned} [P_+(q); \emptyset, g] &\rightarrow [P_+(q); g, \emptyset], & [P_+(q); \emptyset, c] &\rightarrow [M(q, +); \emptyset, c], \\ [P_-(q); g, \emptyset] &\rightarrow [P_-(q); \emptyset, g], & [P_-(q); c, \emptyset] &\rightarrow [M(q, -); \emptyset, c]. \end{aligned} \quad (16)$$

Initialize in $M(q_0, \star)$ with an unmarked blank left stack and the input followed by blanks on the right, marking the right top as the origin (a marked blank for empty input).

For a right move, the primary rule pushes the written symbol left and the transfer removes any tags above the next real right symbol. For a left move, it writes the new right top, removes tags from the left, and moves its next real symbol onto the right. A stay just writes and inserts its tag. Each original instruction adds exactly one tag; at every finite stage there are only finitely many tags to transfer. After j prior machine steps, the next step takes one rule for a stay or two plus the number of transferred tags for a nonzero move, hence at most $j + 3$ rules. If n_k is the rule count after k machine steps, then

$$k \leq n_k \leq k(k-1)/2 + 3k.$$

No intermediate control is terminal. The origin flag recovers the absolute head position: it is k when the mark is the k th real symbol of the left stack and $1 - k$ when it is the k th of the right.

Both rule-cylinder families are disjoint on arbitrary tails. Source controls and tested top symbols distinguish rules. Into $P_+(q)$, primary arrivals have left prefix $b\gamma_{s,a}$ beginning with a real symbol, while loops have a tag there. Distinct tags distinguish the primary arrivals. The analogous argument on the right applies to $P_-(q)$. Exits into $M(q, +)$ or $M(q, -)$ have distinct real right top symbols, while arrivals into $M(q, 0)$ have distinct two-symbol prefixes $b\gamma_{s,a}$. No rule enters a star-tagged main control. These tests also invert each rule; they do not assume that the tails encode an initialized computation.

Let K be the number of controls, set $\lambda = 1/(16(K+1))$, and enumerate nonhalting and halting controls separately from zero. Place their origins at

$$(A_s, 1/4), \quad A_s = \xi_s + (2j_s + 1)\lambda, \quad \xi_s = \begin{cases} 1/8, & s \text{ nonterminal}, \\ 11/16, & s \text{ terminal}. \end{cases}$$

Use odd digits in base $B = 2|\mathcal{A}| + 1$ for the stack alphabet. A prefix w acts on its free tail parameter by $F_w(t) = \sum_j d(w_j)B^{-j} + B^{-|w|}t$, with interval $I(w) = F_w([0, 1])$. Hence each prefix rule gives a positive diagonal map from

$$(A_s + \lambda I(\ell)) \times (1/4 + \lambda I(r)) \quad \text{to} \quad (A_{s'} + \lambda I(\ell')) \times (1/4 + \lambda I(r')).$$

The factors are $B^{|\ell|-|\ell'|}$ and $B^{|r|-|r'|}$. At the first conflicting digit the closed intervals have a positive gap; the full-cylinder argument therefore gives separately disjoint closed rectangle families. It imposes no condition on a source intersecting a target. Constant-tail sums give the rational initial code.

8.4 Fifteen simultaneous phases

We use the rectangles just constructed in Section 8.3, on the unit torus. Both families are separately separated, every half-width is at most $\lambda/2 < 1/2$, and all second coordinates lie in $[1/4, 1/4 + \lambda]$. These are the geometric properties needed for the simultaneous construction; source–target intersections remain allowed. Write the source and target rectangles as $D_i = p_i + [-s_{i1}, s_{i1}] \times [-s_{i2}, s_{i2}]$ and $T_i = q_i + [-t_{i1}, t_{i1}] \times [-t_{i2}, t_{i2}]$. Choose symmetric product cutoffs χ_i^D, χ_i^T on neighborhoods, separately disjoint, with collars smaller than $1/32$. The original supports lie in the common strip $1/4 - 1/32 < y < 1/4 + \lambda + 1/32$; their translates by $\omega = (0, 1/2)$ therefore miss all tracked rectangles and all untranslated supports. For $N > 0$ branches put

$$z_o = 1/2, \quad z_i = 1/4 + \frac{i}{2(N+1)}, \quad d = \kappa = \frac{1}{16(N+1)}, \quad \epsilon = \frac{1}{2} \min_{i,j} \{s_{ij}, t_{ij}\}.$$

Let ζ_i be symmetric about z_i , one near $[z_i - d, z_i + d]$, with support in $(z_i - 2d, z_i + 2d)$. The closures of these supports are disjoint. Define $G_i(z) = (z - z_i)\zeta_i(z)$. Both G_i and G'_i have zero mean, by oddness and differentiation, and are respectively $z - z_i$ and one near the required height interval.

The essential operation changes one coordinate of one sheet. For centers o_i , symmetric rectangle cutoffs χ_i , and factors $\mu_i > 0$, use three successive profiles

$$e_z \sum_i \kappa(y_j - o_{ij})\chi_i(y), \quad e_j \sum_i \frac{\mu_i - 1}{\kappa} G_i(z), \quad -e_z \sum_i \frac{\kappa}{\mu_i} (y_j - o_{ij})\chi_i(y). \quad (17)$$

Starting at height z_i with offset ξ in coordinate j , these stages give height $z_i + \kappa\xi$, horizontal offset $\mu_i\xi$, and height z_i . If $|\xi| \leq 1/2$ and the horizontal segment remains in the cutoff rectangle, the height lies in the linear plateau and no other height profile acts. Every vertical profile has zero mean because its linear factor is odd under reflection about the rectangle center. Every profile is a transverse shear.

Now lift D_i from z_o to z_i using the mean-zero selector $\chi_i^D(y) - \chi_i^D(y - \omega)$. Use (17) first in coordinate one and then coordinate two, with $o_i = p_i$ and $\mu_i = \epsilon/s_{ij}$. This takes six stages and contracts every source to $p_i + [-\epsilon, \epsilon]^2$. Translate by $(q_i - p_i)G'_i(z)$. Next apply the same two triples with $o_i = q_i$, $\chi_i = \chi_i^T$, and $\mu_i = t_{ij}/\epsilon$. Finally lower using the target selector difference. There are $1 + 6 + 1 + 6 + 1 = 15$ stages. Assign them disjoint pulse supports inside $[1/4, 3/4]$.

During contraction the entire projection stays in its source rectangle; during expansion it stays in its target rectangle. Thus all rectangle plateaus used in (17) remain valid, including when the other coordinate has already changed. The translation at height z_i is exactly by $q_i - p_i$. Consequently the period map is

$$(y, z_o) \mapsto (q_i + \text{diag}(t_{i1}/s_{i1}, t_{i2}/s_{i2})(y - p_i), z_o), \quad y \in D_i. \quad (18)$$

For an empty rule list use zero. As in the sequential construction, the normal compensation follows directly from the three shears: in the linear core a triple acts on a small initial height deviation ζ by

$$(\xi, \zeta) \mapsto \left(\mu_i \xi + \frac{\mu_i - 1}{\kappa} \zeta, \frac{\zeta}{\mu_i} \right).$$

Its determinant is one, while its restriction to $\zeta = 0$ multiplies the horizontal coordinate by μ_i . Figure 1 shows this three-shear mechanism on a one-dimensional slice. The sheet statement (18) therefore permits planar area change without claiming that a thick box has that same diagonal action.

Load the fixed label $(1/2, 1/2, 1/2)$ to the rational initial code at height z_o . A profile $G'_o(z)$ supplies a mean-zero horizontal loader if G_o equals $z - 1/2$ near $1/2$ and vanishes near the coordinate

endpoints. Run it once during $[0, 1]$, then repeat the fifteen-stage processor. At $1 + n$ the particle represents rule n by (18). Nonhalt source and target rectangles have first coordinates in $(1/8, 1/4)$; contractions, expansions and the middle translation preserve that band. Loading follows a segment from $1/2$ into that band. Thus a nonhalting run never enters $5/8 < x < 7/8$. Terminal codes are in $(11/16, 13/16)$ and do enter it, including after loading for an initial halt. The finite positive rule counts proved above show that no machine step is left indefinitely unfinished. Direct transverse forcing gives the smooth mean-zero solenoidal force, pressure zero, and periodicity after loading. The repeated processor depends only on the transition table; the input affects only the rational loading code.

9 The equation in material labels

Let u, p be any smooth torus solution constructed above, and let $X(t, a)$ be its material flow. Put $M(t, a) = D_a X(t, a)$ and $P(t, a) = p(t, X(t, a))$. The smooth flow is a diffeomorphism on each finite time interval. Differentiating its equation gives

$$\partial_t M = (D_x u)(t, X)M, \quad \partial_t \det M = (\operatorname{div} u)(t, X) \det M = 0, \quad M(0, a) = I.$$

Thus $\det M = 1$. In particular the observed particle belongs to this volume-preserving flow of the unique solution.

For each component the equation in material labels is

$$\partial_{tt} X_i = -(M^{-T} \nabla_a P)_i + \nu \operatorname{div}_a (M^{-1} M^{-T} \nabla_a \partial_t X_i) + f_i(t, X(t, a)), \quad \det M = 1. \quad (19)$$

Indeed, differentiating $X_t = u(t, X)$ yields the Eulerian material derivative $u_t + (u \cdot \nabla)u$. The chain rule gives $(\nabla_x p) \circ X = M^{-T} \nabla_a P$ and $(\nabla_x u_i) \circ X = M^{-T} \nabla_a X_{i,t}$. To verify the order of the factors in the diffusion term, use a smooth periodic test function φ and volume preservation:

$$\begin{aligned} \int (\Delta_x u_i) \circ X \varphi da &= - \int (\nabla_x u_i) \circ X \cdot M^{-T} \nabla_a \varphi da \\ &= - \int M^{-1} M^{-T} \nabla_a X_{i,t} \cdot \nabla_a \varphi da. \end{aligned}$$

Periodic integration by parts proves the asserted diffusion operator; smoothness upgrades this test-function identity to a pointwise one. The shear constructions have $p = 0$ and therefore $P = 0$. Their prescribed open-set events consequently observe the material evolution governed by (19) itself.

A A finite interpreter in the two-sided tape model

Before serialization, delete all outgoing rows from designated halt states. As in the recorder constructions, complete every missing state–symbol pair at a nonhalting state by an unchanged-symbol, zero-displacement move to a fresh halt state with no outgoing rows. This finite normalization preserves the halting question, including an initially halted machine. In the resulting table, failure to find a matching row is equivalent to being in a halt state.

Give the interpreted states and symbols positive integer indices, with blank index one. Write an index m as a block of m ones followed by a separator. Use distinct syntax symbols for begin, row-end, table-start, table-end, and the three head motions. Serialize a machine and input by its initial state, its input-symbol fields in reverse word order, and its finite list of transition rows. A

row contains current state, scanned symbol, next state, written symbol, and motion. Correctness is required on these valid finite descriptions.

First construct a fixed five-tape interpreter. The tapes hold the unchanged description, a current-state record, a left-cell stack, a right-cell stack, and a scratch record. Each work tape has a sentinel at cell zero. A record is a finite unary block immediately to its right, with its head returned to the sentinel between uses. A stack consists of finitely many unary symbol blocks to the right of its sentinel, written bottom to top; each block-end symbol also carries an origin bit. The head rests at its top block-end, or at its sentinel when empty. Below the stored blocks the logical stack consists of unmarked blanks.

The needed routines have finite control and terminating scans. To copy a field, erase the old record, return to its sentinel, and copy the field's ones while scanning it. To compare a field and a record, scan their ones in parallel, remember whether they end together, and return the record head. To push, copy the scratch unary block after the top and append the chosen origin-bit block-end. To pop, retain the top origin bit in control, erase its end, and traverse its ones backward, copying one to scratch for every erased one until the previous end or sentinel. An empty stack instead supplies blank index one and origin bit zero. All these scans stop on explicitly present finite delimiters.

Initialize the state record from the description. Push the reversed input fields onto the right stack, then mark its top block as origin; for an empty input push a marked blank. Leave the left stack empty. In each iteration pop the right top into scratch, remembering its bit. Search the table for a row matching the state and scratch records. On a match update the state and scratch to the next state and written symbol. A right move pushes the written symbol onto the left; a stay pushes it back onto the right. A left move first pushes the written symbol right, then pops the left (supplying an implicit blank if empty) and pushes that symbol right. Every push retains the appropriate origin bit. Return the description head to table-start and repeat. If no row matches, restore the popped symbol and halt; by the normalization above, this also handles a halt on the first iteration. The two logical stacks are therefore exactly the interpreted tape halves, and the origin bit recovers its absolute head position at every iteration.

These routines compile to a finite five-tape table: instruction location, comparison outcome, origin bit, and chosen direction have finitely many values; unbounded interpreted indices stay in unary on tape. Each character test, write, copy, or move is an ordinary finite transition, and the finitely many call sites can be inlined. Malformed descriptions may receive arbitrary defaults.

Finally encode the five aligned tracks on one two-sided tape between left and right fences, with one head bit per track in the product alphabet. At a simulated step, sweep the finite active interval to collect all five scanned symbols into finite control. Choose the transition before changing any track. Then sweep to each head bit in turn, make that track's write and move, preserving every other track. If a head crosses a fence, replace that fence by a blank tuple with the new head bit and put the new fence one cell farther out. Return to the left fence between sweeps. Only finitely many tracks, symbols, and pending transition choices are involved, so these are again finite rules. Every sweep terminates, and all data and head bits match the five-tape machine at step boundaries. Halt exactly when it halts. This yields a fixed universal single-tape table. Serialization and fluid compilation use only the finite description and input, and never run the interpreted computation in advance.

The same fence construction also converts any fixed finite number of tapes to one before applying a recorder: collect the finitely many head symbols, move the fences outward when needed, update one track at a time, and return to the left fence. If a one-sided tape convention is desired, add a preserved bit at its left endpoint and implement that convention's left-end behavior in the marked-symbol transitions. Both reductions preserve the initialized halting question without changing the geometric arguments.

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